

Solving the Navier-Stokes Equation: A Machine Learning Approach

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Background and Motivation

Computational Fluid Dynamics (CFD) is essential for physical systems

- Weather and climate models
- Aerodynamic designs
- Biomedical Flows (blood flow simulations)

Governed by Navier Stokes equations, CFD simulations face major challenges

- Extremely high computational cost
- Need to resolve smallest flow features for accuracy
- Trade-off between accuracy and simulation speed

Navier-Stokes equations

Incompressible fluids are modeled by the Navier Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} = -\nabla \cdot (\mathbf{u} \otimes \mathbf{u}) + \frac{1}{Re} \nabla^2 \mathbf{u} - \frac{1}{\rho} \nabla p + \mathbf{f} \quad (1a)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (1b)$$

where $\mathbf{u}$ is the velocity field, $\mathbf{f}$ the external forcing, and $\otimes$ denotes the tensor (outer) product. The density ρ is constant, and The Reynolds number Re characterizes the ratio of inertial to viscous forces in the fluid.

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Challenges

Computational Complexity

- The Reynolds number (Re) indicates the balance between convection and diffusion.
- High Re ($\gg 1$) means convection dominates, leading to turbulence and increased complexity.
- Resolving turbulent flows requires computational cost scaling as Re^3 .

Aim

- Aim to accelerate DNS without compromising accuracy and generalization.
- Use ML models to enhance coarse-grid CFD solvers via super-resolution of missing details.

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Machine Learning Model

- We propose ML-augmented solvers to accelerate Direct Numerical Simulation (DNS) by enhancing coarse-grid Computational Fluid Dynamics (CFD) solvers without sacrificing accuracy.
- Two ML components are introduced:
 - **Learned Interpolation (LI)**: Neural networks predict interpolation coefficients to compute convective fluxes $\mathbf{u} \cdot \nabla \mathbf{u}$ more accurately than standard polynomial methods.
 - **Learned Correction (LC)**: A residual model where the neural network predicts corrections as:

$$\mathbf{u}_t = \mathbf{u} + \text{LC}(\mathbf{u})$$

where $\mathbf{u}$ is the coarse solver output, and $\text{LC}(\mathbf{u})$ is the learned correction.

- The neural networks use convolution layers for local pattern detection and are integrated into a simulation that updates values over time on a staggered grid.

Machine Learning Model

- The complete model is trained end-to-end via supervised learning to minimize the cumulative mean-squared error (MSE) loss over time steps:

$$L(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^T \text{MSE}(\mathbf{u}_{\text{exact}}(t_i), \mathbf{u}_{\text{pred}}(t_i))$$

where $\mathbf{u}_{\text{exact}}$ is from high-resolution DNS and $\mathbf{u}_{\text{pred}}$ is the model prediction.

- The learned model improves the accuracy of convection approximation and enforces physics-based inductive biases like momentum conservation and incompressibility, enabling stable long-term predictions with reduced computational cost.

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Problem statement

We study the two-dimensional incompressible Navier-Stokes equations for Newtonian fluid flow in a channel along the x - z plane. The governing equations are:

Continuity:

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0 \quad (1)$$

Momentum:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (2)$$

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (3)$$

Domain: $x \in [0, L]$, $z \in [0, H]$

Problem statement

Initial Conditions:

$$u(x, z, 0) = u_0(x, z), \quad w(x, z, 0) = w_0(x, z)$$

Boundary Conditions:

- No-slip at walls ($z = 0, H$): $u = w = 0$
- Inlet ($x = 0$): $u = U_{\text{in}}(z)$, $w = 0$
- Outlet ($x = L$): $\frac{\partial u}{\partial x} = \frac{\partial w}{\partial x} = 0$ or $p = p_{\text{out}}$

Objective: Solve for (u, w, p) using machine learning methods satisfying physics-based constraints.

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Thank you for listening !

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